
Aryama Vinay Murthy:

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📍 Bengaluru, India | 🎓 B.Tech in Computer Science, IIIT Hyderabad (2024–2028)
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PROFILE SUMMARY:

- **B.Tech Student at IIIT Hyderabad:** First-year student with a strong passion for artificial intelligence, machine learning, and product innovation.
 - **Innovative Product Builder:** Adept at ideation, planning, and executing projects that blend broad technical knowledge with real-world applications.
 - **Hackathon Achiever:** Winner at multiple hackathons, including the Felicity Hackathon, Agri-India Hackathon and HackIIIT, building solutions such as a streamlined virtual lab creator, AI-driven meme maker and agricultural system for weed and disease detection and smart watering system.
 - **AI/ML Enthusiast:** Passionate about exploring machine learning techniques, neural networks, transformer models, fine-tuning techniques, and physical AI with robotics , with a strong interest in building and applying AI-driven solutions.
 - **AI Coding Expert:** Skilled in using AI coding tools, copilots, and prompt engineering to boost development efficiency and enhance project outcomes.
 - **Collaborative Leader:** Brings excellent people skills and a visionary approach to project ideation, planning and execution
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ACHIEVEMENTS:

- **Felicity Hackathon:** 1st Place for One-Click-Labs amongst a 2000 students participating all over India(awarded 75,000)
 - **HackIIIT Hackathon:** Won an award with the AI Meme Maker project for making the most fun product in the hackathon.
 - **Agri-India Hackathon:** Winner in Mechanised Farming(awarded 1 lakh) amongst multiple participants(this hackathon was open for all individuals and companies)
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Education:

Education	Institution	Years	Results
B.Tech in Computer Science	International Institute of Information Technology, Hyderabad (IIIT-H)	2024 – 2028	Currently ongoing..
Pre-University (PU)	RV PU College	2022 – 2024	- 96.33% in 12th Board Exam - 99.9347%ile (AIR 1205) in JEE Mains 2024 - AIR 3576 in JEE Advanced 2024 - State Topper in Physics Olympiad
Schooling (CBSE)	Sri Kumarans Children's Home	2012 – 2022	- 94% in 10th Board Exam

TECHNICAL SKILLS

Programming Languages

- Python, C, C++, JavaScript, Bash scripting, SQL, TypeScript, HTML, x86 Assembly.

AI & Machine Learning

- **ML/DL Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib.
- **AI Concepts:** Fine-Tuning (LoRA, QLoRA), RAG, Model Quantization, Multi-Agent Systems, Attention Mechanisms, CNNs, RNNs, LSTMs, GRUs
- **LLM Frameworks:** LangChain, LangGraph, Vector Stores (ChromaDB, FAISS), Document Loaders, Text Splitters, Tool calling.
- **Mathematical Foundations:** Discrete Mathematics, Linear Algebra, Statistics and Probability, Calculus.

Software Development

- **Frontend:** React, Next.js, React Native (Expo) (Using AI tools to create the frontend in a rapid and seamless manner)
- **Backend:** FastAPI, Flask, Express.js
- **Databases:** PostgreSQL, MongoDB, Firebase, Supabase
- **Development Efficiency:** AI-assisted coding, Git version control , Docker,Rapid Prototyping with software and microcontroller and IOT systems.

IoT & Embedded Systems

- **Platforms:** ESP32, Raspberry Pi, Arduino
- **Protocols:** MQTT, REST APIs, Bluetooth/Wi-Fi Sensor Integration
- **Applications:** Motion Tracking, Sensor Data Processing, Edge AI, Cloud processing for computer vision.

PROJECTS & EXPERIENCE:

Robotics – Agri-India Hackathon

Team Lead and hackathon winner

Overview: Led a team of three in ideating and developing an agricultural robot prototype for weed removal and disease detection using machine learning.

Role: Directed the project vision, technical design, and business planning. Collaborated on building the robot and the end to end IOT system.

Outcome: Awarded in the Mechanised Farming category at the Agri-India Hackathon (Government of India).

Technologies: Raspberry Pi, Arduino, IoT integration, Tensorflow, OpenCV.

Felicity Virtual Labs Hackathon – One-Click-Labs(team of 2)

Won 1st place out of 2000+ participants

Hackathon Winner and lead

- **Overview:** Developed One-Click-Labs, a solution that streamlines the creation of virtual lab experiments and study material for professors.
 - **Role:** Led the ideation, planning, and design of the AI pipeline architecture.
Built the entire website with next js, fast API and mongoDB.
 - **Outcome:** Achieved 1st Place in the Felicity Hackathon sponsored by VLABS and VLEAD.
 - **Technologies:** LangChain, LangGraph, FastAPI, Next.js, MongoDB, RAG.
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HackIIIT Hackathon – AI Meme Maker(using monty cloud's moyaAI platform)(Team of 5)

Hackathon Winner

Overview: Developed an AI-powered meme maker—a Discord bot that generates memes using large language models (LLMs) and the MoyaAI framework, integrated with RAG techniques and Chroma DB.

Role: Helped in Ideation and creation of the AI component design and implementation, ensuring seamless integration with the Discord API.

Note: The hackathon was conducted by OSDG, IIIT Hyderabad in collaboration with Monty Cloud.

Outcome: We won the hackathon with the emphasis on creating the most fun product in the hackathon.

Technologies: Discord API, LLMs, MoyaAI framework, RAG, Chroma DB

Automatic navigating robot in a closed environment using a camera

Project

Overview: Built a vision-guided autonomous robot in a team of 3 using a top-mounted camera and ESP32.

Enabled real-time obstacle avoidance and point-to-point navigation using image segmentation , path planning and path following.

Technologies: OpenCV, Image Segmentation, ESP32, CoAP Protocol, Path Following, NumPy

Fine-Tuning Florence 2 for OCR

Participated in a **Kaggle OCR competition**, fine-tuning **Florence 2** for handwriting recognition.

Technologies: Hugging Face Transformers, PyTorch, Bitsandbytes, Pandas

AI-Based IoT Exercise Analysis

Designed a real-time exercise tracking system using gyroscope & accelerometer data from ESP32.

Technologies: FastAPI, React Native, Pandas, Scikit-learn, DTW

Courses taken and Learning experiences:

Online Courses:

- **Complete Generative AI Course With Langchain and Huggingface on Udemey**
Completed a comprehensive course by Krish Naik, focusing on building, deploying, and optimizing Generative AI models using Langchain and Huggingface. Gained hands-on experience in creating real-world AI systems.
- **Python for Data Science and Machine Learning Bootcamp on Udemey**
Completed a course by Jose Portilla, covering data analysis, visualization, and machine learning algorithms using Python. Developed proficiency in libraries such as Pandas, NumPy, and Scikit-learn.

College Courses:

- **Discrete Mathematics, Real Analysis, Linear Algebra, Data Structures & Algorithms, Computer Systems Organisation, Software Systems, IoT.**

Others:

Just Robotics Education: Theoretical and Hands-on experience in robotics, electronics, and IoT.

Hobbies & Interests

- **Lawn Tennis:** Played for the college team.
 - **Chess:** FIDE-rated player with a 1464 rating.
 - **Music:** Plays Carnatic classical flute.
 - **Photography:** Has traveled across India and the world, capturing moments through the lens.
 - **Finance & Current Affairs:** Interested in staying informed and learning about finance and following current affairs.
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Additional Experience & Soft Skills

- **AI Coding & Prompt Engineering:** Skilled in leveraging AI coding tools, copilots, and prompt engineering to accelerate development and problem-solving, as demonstrated through rapid prototyping in hackathons.
- **Team Collaboration:** Experienced in leading and working with multidisciplinary teams to ideate, plan, and execute projects efficiently. Proficient in collaborative development using Git and other version control tools.
- **Visionary Innovative Thinking & Entrepreneurial Mindset:** Passionate about translating AI innovations into scalable, future-ready solutions. Focused on building software with long-term impact and adaptability.
- **Continuous Learning & Adaptability:** Committed to staying at the forefront of emerging technologies. Quick to learn, adapt, and apply new concepts, continuously taking on challenges to develop impactful AI-driven products.